PlugPlan SG Demo Script (Live data throughout)

SC2006 · SCE3-04 | Main flow about 12–15 min | Button names match the UI exactly | **Contains test passwords: do not commit to GitHub or show on the projector**

0 · Before you start

Item	Details
URL	`https://plugplan-sg.vercel.app` (backup: run locally with `cd "PlugPlan SG"` → `pnpm dev` → `http://127.0.0.1:3000`)
Window 1 **Driver**	Chrome, normal window
Window 2 **Moderator**	Edge (or a Chrome Incognito window). **Two normal tabs in the same browser share one login, so that won't work.**
Window size	Maximise the browser window. In a narrow window the sidebar shows icons only, with no labels.
Upload image	`Lab 3/demo-support/demo-evidence.png` (for the wrong-format demo, use `invalid-upload.svg`)
OneMap token	The current token expires at **16:05 SGT on 9 Nov 2026**. If the demo is after that, first run: `node "PlugPlan SG/scripts/renew-onemap-token.mjs" --vercel`

Accounts

Role	Email	Password
Driver A main	`driver-20261015@plugplan.test`	`RTfDfJYgn8u2NtSCfjkIi5MSA1!`
Moderator	`moderator-20261015@plugplan.test`	`NguT38LO-75D0KjdNkAM95M2A1!`
Driver B spare	`other-20261015@plugplan.test`	`b6uKwbV8fXQ9Bc1fPPntsbBMA1!`

Keep the passwords in a Notepad file and paste them. Don't type them live.

Pre-demo checklist (do all of these)

1. Sign in as Driver A in one window and as Moderator in the other. Confirm both work.
2. Moderator → **Report review** → **Refresh**. The **Pending** and **Verified** filter counts should both be 0. If not, open each report: **Review** → Decision **Resolve** → Reason `Pre-demo cleanup.` → **Record decision** → click **OK** in the browser dialog.
3. In the Driver A window, **do one full dry run** with the UC-03 inputs below. Note the names of the first and second results and check there are at least 2 options. This also warms the server cache.
4. Sign out of the Driver window and leave it on the sign-in page. The demo starts from sign-in.

> Live data changes in real time. Station names, prices and availability can be different on every run. The names in this script (TIARA, THE JTC SUMMIT) are from the 6 November rehearsal. Say "the first recommendation" and "the second recommendation" instead of memorising names.

1 · Main script

UC-01 Sign in Driver window

1. (Optional) Click **Create an account** to show sign-up: public sign-up creates Driver accounts only, and the password needs at least 10 characters with a letter and a number. **Don't submit.** Click **Sign in** to go back.
2. Enter the Driver A email with a wrong password, `wrongpass123` → **Sign in**.
3. Enter the correct password → **Sign in**.

> Wrong password: a generic error that doesn't reveal whether the account exists. After signing in, the sidebar shows Plan a journey / Saved plans / Your vehicle / Your reports.

> *"PlugPlan SG helps EV drivers in Singapore find a charging stop that fits their trip, using live data from LTA DataMall and OneMap. Supabase Auth handles sign-in, and the error message doesn't reveal whether an account exists. There are two roles, Driver and Moderator, and the server checks permissions."*

UC-02 Vehicle profile Driver

1. Sidebar → **Your vehicle**.
2. Check or enter: Vehicle name `Demo EV`, Usable battery capacity `60`, Energy consumption `17.5`, Reserve state of charge `20`, Compatible plugs **Type 2** + **CCS2 / Combo 2**, AC `11`, DC `120`. In the right panel: Default maximum detour `15`, Minimum charger power `0`, Maximum walk `0`.
3. Click **Save vehicle**.

> The vehicle saves and the app moves to Plan your journey. The bottom right shows "PLANNING WITH Demo EV · 60 kWh".

> *"Each driver keeps one vehicle profile. Battery size, consumption, plug types and maximum power all go into the calculation. A 20% reserve means we always arrive with at least 20% left."*

UC-03 Create a charging plan (Live) Driver

1. Click **Live data** in the top right. It must not be Demo fixtures, and there should be no yellow Demo banner.
2. In Starting from, type `Nanyang Technological University`. **Wait for the dropdown and click the first result** (NANYANG TECHNOLOGICAL UNIVERSITY · 94 NANYANG CRESCENT). A green ✓ and "Location selected" appear.
3. In Going to, type `Marina Bay Sands` and click the first result, **MARINA BAY SANDS**.
4. Set Current battery to `18` (the default may differ). Leave Departure time at the current time. Available charging time `45`, Maximum detour `15`, Ranking strategy `Balanced`, Charging target `Just enough for this journey`.
5. Click **Find charging options** and **wait 3–5 seconds**. Say the first sentence of the script below while you wait.
6. Expand **Assumptions & considerations** on the first card, then scroll down and click **See why they didn't fit**.

> Version 1 · create, labelled **Live · Source: LTA DataMall** with Observed / Retrieved times. Usually 2–3 options. In rehearsal: ① TIARA (AC 11 kW, about 36 min, S$3.19, arrives at 11.4% → target 21.2%) and ② THE JTC SUMMIT (DC 120 kW, about 3 min, S$4.93). A note says only 8 candidate stations in the route corridor were checked.

> *"The app is now asking OneMap for routes and LTA DataMall for charger locations and status across Singapore, in real time. Each result shows its data source and when it was retrieved. We don't tell you to charge to 100%. We work out just enough to reach the destination with the 20% reserve. Each card shows the estimated charging time, cost, detour and why it's recommended. At the bottom you can see which stations were excluded and why, for example an incompatible plug, too long a detour or no free connector right now."*

If there's only 1 option: **Edit journey** → Maximum detour `20` → **Recalculate plan**, then carry on.

UC-04 Compare and choose Driver

1. Click **Compare options**. With nothing ticked, it compares all of them.
2. In the first-ranked column, click **Select charger**.
3. Click **Back to options**.
4. Under the map on the right, click the green **Navigate via …** button (or **Navigate via this stop in Google Maps** on any card). Google Maps opens in a new tab. Switch back to the PlugPlan tab afterwards.

> Every option is compared on the same metrics: cost, charging time, detour, arrival battery, target, total time and score. After you choose, the toast reads "… selected. This does not reserve the charger." and the card button changes to **Selected**. The version number stays the same. Google Maps shows a driving route: origin → the chosen charging station (as a stop) → destination.

> *"For example, the first one is cheaper but charges slowly, and the second is a fast charger but costs more. Choosing a stop only saves a preference. It doesn't create a new version, and it isn't a booking. The app doesn't reserve chargers or take payments. Once you've chosen, one click takes you to Google Maps to navigate this plan, with the charging stop on the way."*

UC-05 Reconfigure and versions Driver

1. Click the strategy tab **Fastest journey**.
2. Switch the **Plan version** dropdown to **Version 1 · create**. Point out that the old result is still there and **Choose this stop** is disabled. Then switch back to the latest version.
3. Click **Balanced** to switch back.

> "Strategy changed. A new version was saved." This creates Version 2. In rehearsal, THE JTC SUMMIT moved to first, but the order depends on the data at the time. The data label may change to **Cached**. Back on Balanced: Version 3, and the stop you chose is still Selected.

> *"Every recalculation adds a new version. Old versions keep their inputs and results. You can view them but not change them. The Cached label means the server reused data for the same route from the last 10 minutes. It doesn't pretend the data is new, and it shows the real timestamps."*

Optional extra: **Edit journey** → Available charging time `60` → **Recalculate plan**. Or click **Cancel changes** to show that cancelling doesn't create a version.

UC-07 Report an issue Driver

1. On the **Selected** card, click **Report an issue at this charger**.
2. What happened: `Suspected faulty connector`.
3. A few helpful details: `Demo report (test only): screen is off and the connector does not start charging.`
4. Photo (optional) → **Choose a photo** → `demo-evidence.png`.
5. Click **Confirm & submit report**.

> The app moves to Your reports. The new report is **Pending**, has a reference (#xxxxxxxx) and expires in 24 hours. You can click **View** → **View evidence image** to show the photo.

> *"Drivers can report a charger problem and attach a photo. Only JPEG, PNG or WebP up to 5 MB is accepted. The form shows the operator's name to remind drivers that repairs still go through the operator's official channel. The report starts as Pending. One unreviewed report never removes a station. A Moderator has to review it first."*

UC-08 Review the report Moderator window

1. Switch to the Moderator window → **Report review** → **Refresh**.
2. Open the newest Pending report, for the station you just reported, with **Review**. Point out the details, the photo, related reports at this station and the **Official status** (what LTA reports for this connector).
3. Decision: `Verify`. Reason: `Demo review: evidence checked, fault confirmed.`
4. Click **Record decision**.

> The message reads "Decision recorded in the audit history." The status changes to **Verified**, and Decision history shows a new entry with the time and the Moderator.

> *"The Moderator sees the driver's evidence next to LTA's official status. Every decision needs a reason and goes into the audit history. The other actions are Reject, Merge, Resolve and Expire. Each update checks the revision number, so if two moderators decide on the same report at once, the second one gets a 409 conflict."*

UC-06 Refresh and pick an alternative Driver window

1. Switch back to the Driver window → **Saved plans** → **Open plan** on the first plan.
2. Click **Refresh data** and wait 3–5 seconds.
3. On the suggested alternative's card, click **Choose this stop**.

> A new version (refresh) appears, and the reported station is gone. The message reads "Your saved selection (...) was cleared because ... an active verified faulty-station mark ... Next alternative: ... Select it to continue."

> *"Refreshing fetches live data again and filters out stations with a verified fault. The old choice is cleared and the app suggests the next option, but the driver makes the final pick. The app never switches it automatically, and older versions aren't changed."*

Wrap-up Reset and sign out

1. Moderator: open the same report with **Review** → Decision `Resolve` → Reason `Demo cleanup: connector repaired.` → **Record decision** → click **OK** in the dialog.
2. Driver: **Refresh data**. The station is back in the list.
3. Click **Sign out** (top right) in both windows.

> *"Once the fault is resolved, the station is back in the results on the next refresh. That's the full loop: the driver plans, compares and reports, and the Moderator reviews."*

> Always do step 1. You reported a real charging station, and it stays filtered out until the report is resolved.

2 · Extra demos if asked (each under 1 min, in Live data)

Show	Steps (Driver)	Expected
Ambiguous place (UC-03)	In Starting from, type `Orchard`	Several places in the dropdown. You have to click one to select it
No charge needed (UC-03)	Same journey, Current battery `100` → Find charging options	"You already have enough charge."
Direct navigation (UC-04)	After the Current battery `100` result, click **Navigate direct route** under the map	Google Maps opens origin → destination with no stop
No suitable stop (UC-03)	Same journey, Maximum detour `0` → Find charging options	"No suitable stop this time." with exclusion reasons
Manual target (UC-03)	Same journey → **Set my own target** → `60` → Find charging options	Cards show both the automatic and manual targets. Slow chargers may be excluded
Vehicle validation (UC-02)	Your vehicle → untick all Compatible plugs → Save vehicle	Field error and nothing is saved. Re-tick Type 2 + CCS2
Unsaved changes (UC-05)	Edit journey → change a field → click Saved plans	Dialog: "Leave without saving your changes?"
Wrong image type (UC-07)	In the report form, choose `invalid-upload.svg` → submit	Rejected, and the text you typed is kept
Duplicate report (UC-07)	Submit again for the same station and category while that station's report is still Pending or Verified	"An active report already exists". No new report is created
Reject / Expire (UC-08)	Report another station as `Blocked charging bay` (no photo) → Moderator chooses Reject (or Expire), adds a reason and clicks OK	Status Rejected / Expired, with an audit entry
Fixed-data mode (UC-03)	**Demo fixtures** → **Use example journey** → departure 20:00 → Find charging options	3 synthetic stations (Demo Fast Charge S$4.37, etc.), labelled DEMO_FIXTURE

Each account can submit at most 5 reports per hour, and a rejected duplicate also counts. If you run out during rehearsal, switch to Driver B. Reject, Merge, Resolve and Expire open a browser confirm dialog, so click OK. Resolve every station you reported before you finish.

3 · Likely questions (one-line answers)

Question	Answer
Why not charge to 100%?	The target is just enough to reach the destination plus the 20% reserve, which saves time. Drivers can set a higher target with Set my own target.
Where does the data come from, and how fresh is it?	LTA DataMall provides stations, connectors, status and prices. OneMap provides place search and driving routes. Cards show observed and retrieved times, with a warning once data is more than 10 minutes old.
What do Live / Cached / Stale mean?	Live: just fetched. Cached: reused from the server cache within 10 minutes. Stale: an upstream call failed, so older cached data is used with a warning.
Why only 8 candidate stations?	To respond within a few seconds, we only calculate exact routes for the nearest compatible stations along the route. The UI says the result isn't guaranteed to be the best in all of Singapore.
Why don't Pending reports exclude a station?	To guard against mistaken or malicious reports. Only Verified, unexpired fault reports affect the filter.
Why doesn't choosing a stop create a version?	A selection is a preference that can change. Only recalculations (strategy change, edit, refresh) create new, immutable versions.
Can it navigate?	Yes. The button builds a Google Maps directions link from the origin, charging stop and destination coordinates, and opens it in a new tab (the Google Maps app on a phone). We don't do turn-by-turn navigation ourselves.
Can I book a charger?	No. The app gives recommendations with their reasons. It doesn't handle bookings or payments.
How are permissions enforced?	The server checks the role stored in the database. A Driver calling the moderation API gets 403, and opening someone else's plan returns 404.
Tech stack?	Next.js and TypeScript, with Supabase for Auth, Postgres and Storage, deployed on Vercel in the Singapore region. Data comes from LTA DataMall and OneMap.

4 · If something goes wrong

Symptom	Fix
"A live data request timed out."	Click again. The second attempt uses the server cache and usually succeeds.
Place search shows nothing	The OneMap token has expired. Run `node "PlugPlan SG/scripts/renew-onemap-token.mjs" --vercel` (about 1 minute).
Only 1 option, so nothing to compare	Edit journey → Maximum detour 20 → Recalculate plan.
Report submission hits the rate limit	Submit from Driver B instead.
Report says "already exists"	That station already has an open report. Have the Moderator review that one instead.
Live data is completely down during the demo	Explain why, switch to **Demo fixtures** → Use example journey → departure 20:00, and finish the main flow on fixed data (Demo Fast Charge / Demo Reported Fault / Demo Value Charge).
The live site is down	Use the local app at `http://127.0.0.1:3000`. It uses the same database and the same accounts.